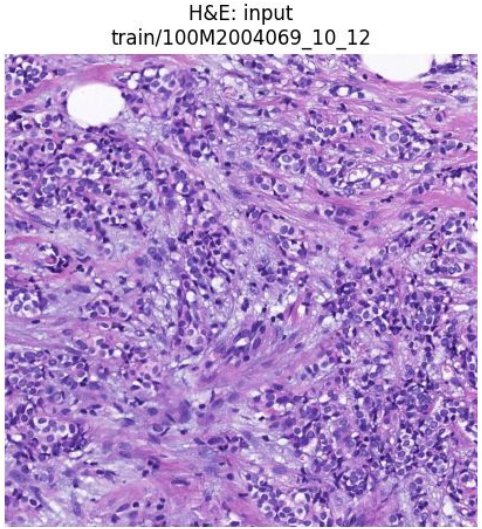
NEUROML FINAL PROJECT

Flow Matching for Virtual IHC Staining

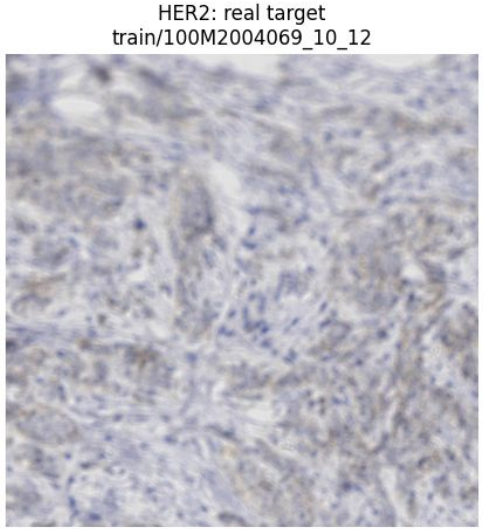
H&E-conditioned generation of HER2 immunohistochemistry

Andrey Galichin
Vasilii Gorelov
Urazmetova Margarita

Virtual staining predicts molecular information from routine H&E



H&E



HER2 IHC

Clinical motivation

H&E reveals tissue morphology but does not directly identify biomarker expression.

HER2 IHC highlights membrane-associated protein expression used in breast cancer assessment.

A virtual stain could reduce additional laboratory work and preserve limited tissue, while remaining a research tool rather than a diagnostic replacement.

Team

Staining Flow Swimmers

Vasilii Gorelov

FM methodology & modeling

Baseline CFM pipeline

Alternative FM methods

Architecture and quality

Andrey Galichin

FM methodology & modeling

HER2match implementation

Dataset preparation

Preprocessing and splits

Margarita Urazmetova

MIST & experiments

Metrics and baselines

Sampling and ablations

Quality and failure analysis

Problem Statement: Virtual H&E-to-HER2 Staining

Problem: H&E shows tissue morphology but does not directly reveal HER2 protein expression. Laboratory IHC provides this information but requires additional tissue, time, and resources.

Task: Generate a virtual HER2 IHC image from an H&E image while preserving tissue morphology and reproducing spatially correct HER2 staining.

Criterion	Requirement
Paired translation fidelity	Agreement with the corresponding real HER2 image
Target-distribution fidelity	Agreement with the distribution of real HER2 IHC
Biomarker fidelity	Correct intensity and localization of HER2 staining
Quality–efficiency trade-off	Generation quality relative to computational cost

Existing approaches

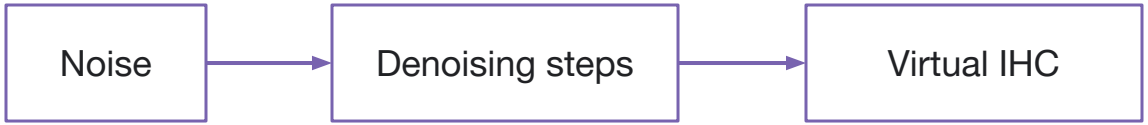
GAN translation

One generator pass



Conditional diffusion

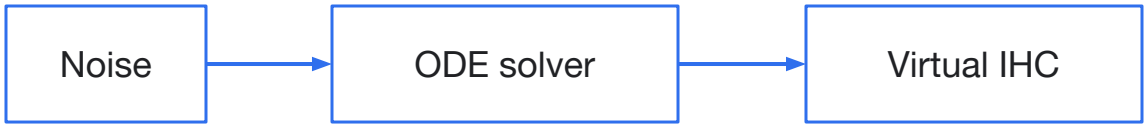
Iterative denoising



H&E condition at every network evaluation

Flow Matching

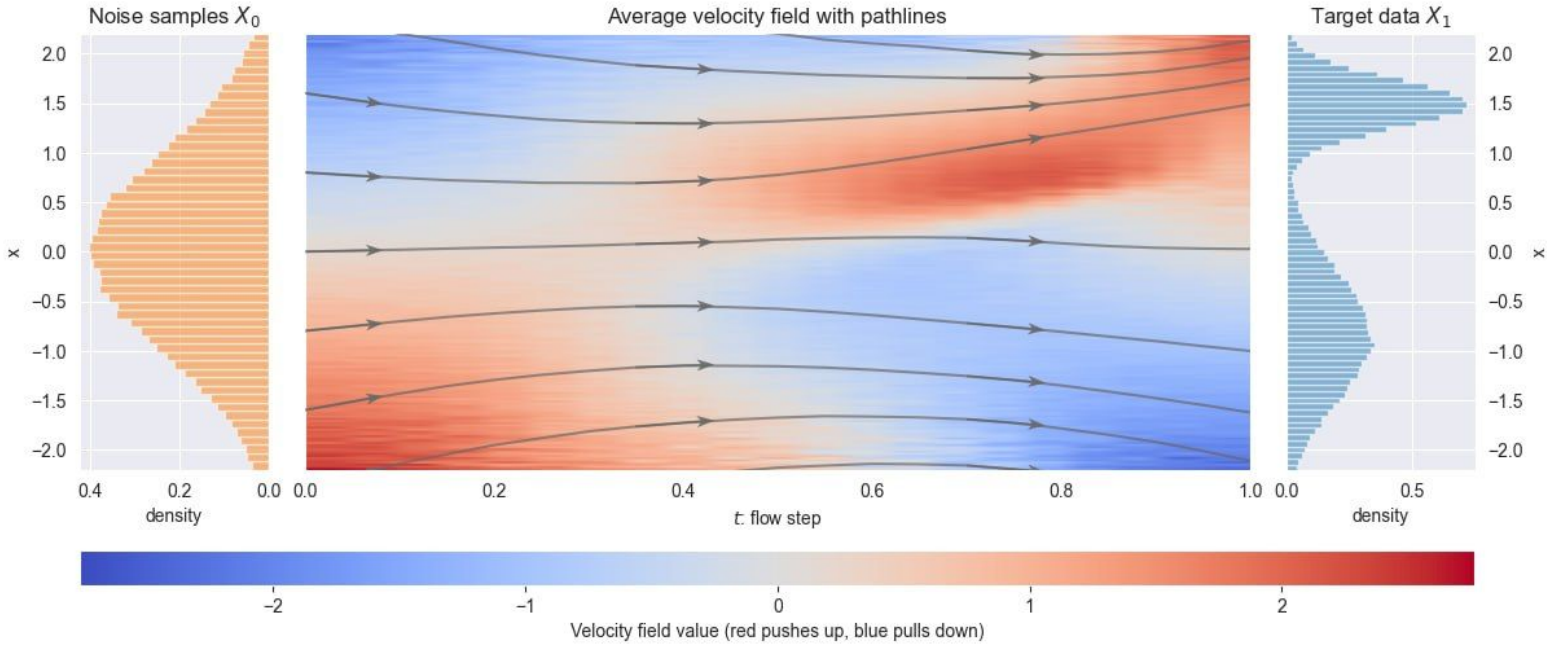
Conditional velocity and ODE



H&E condition at every network evaluation

Research question **How do transport paths and sampling affect structure and staining?**

Conditional Flow Matching learns how to move noise toward HER2



Training path

$x(t) = (1-t)z + ty$

Regression target

$v^*(x,t) = y - z$

Condition

H&E remains fixed during ODE sampling

Two datasets

MIST-HER2

PRIMARY BENCHMARK

3,805 train | 784 validation | 1,053 test
Adjacent tissue sections can contain local inconsistencies. Published baselines enable comparison with prior work.

HER2match

ALIGNMENT STUDY

11,610 train | 3,582 validation | 5,980 test
21,172 paired tiles from the same tissue sections. Better registration supports a controlled test of alignment effects.

MIST-HER2: testing the baseline

01	**Baseline CFM**	Different H&E inputs produced similar textures. Tissue structure was poorly preserved.
02	**Path and crop size**	VP-CFM performed worse. Smaller crops raised SSIM, but worsened FID and KID.
03	**Optimal transport**	OT did not visibly fix the weak dependence on H&E.

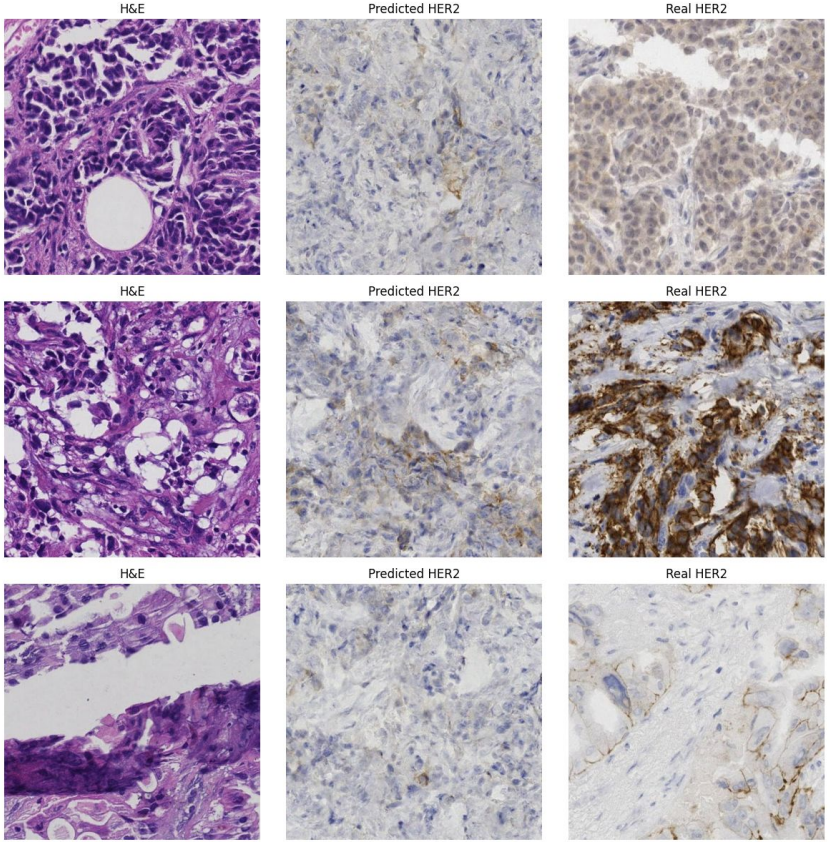
Main problem: the model did not follow the input tissue closely enough.

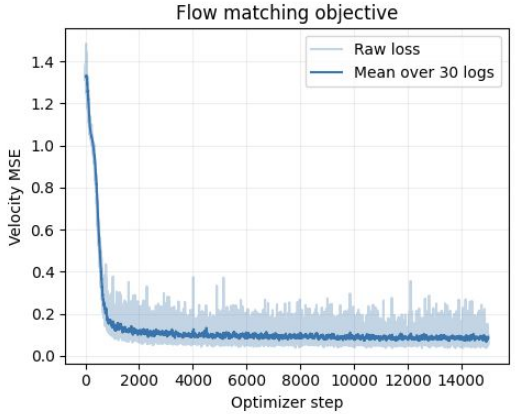
MIST-HER2: stronger guidance

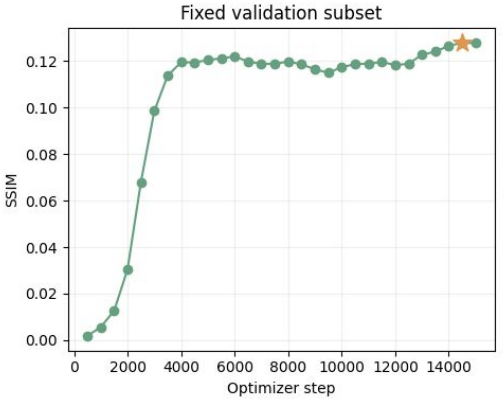
04	**Pathology features**	Added Phikon features and structure/stain losses. Higher SSIM did not ensure better staining.
05	**Direct flow from H&E**	Outputs followed H&E more closely, but looked like masks. Poorly aligned pairs looked worse.
06	**Next: HER2match**	Moved to better-aligned pairs to investigate the role of alignment.

Remaining problem: realistic HER2 staining and correct marker placement.

Baseline MIST-HER2

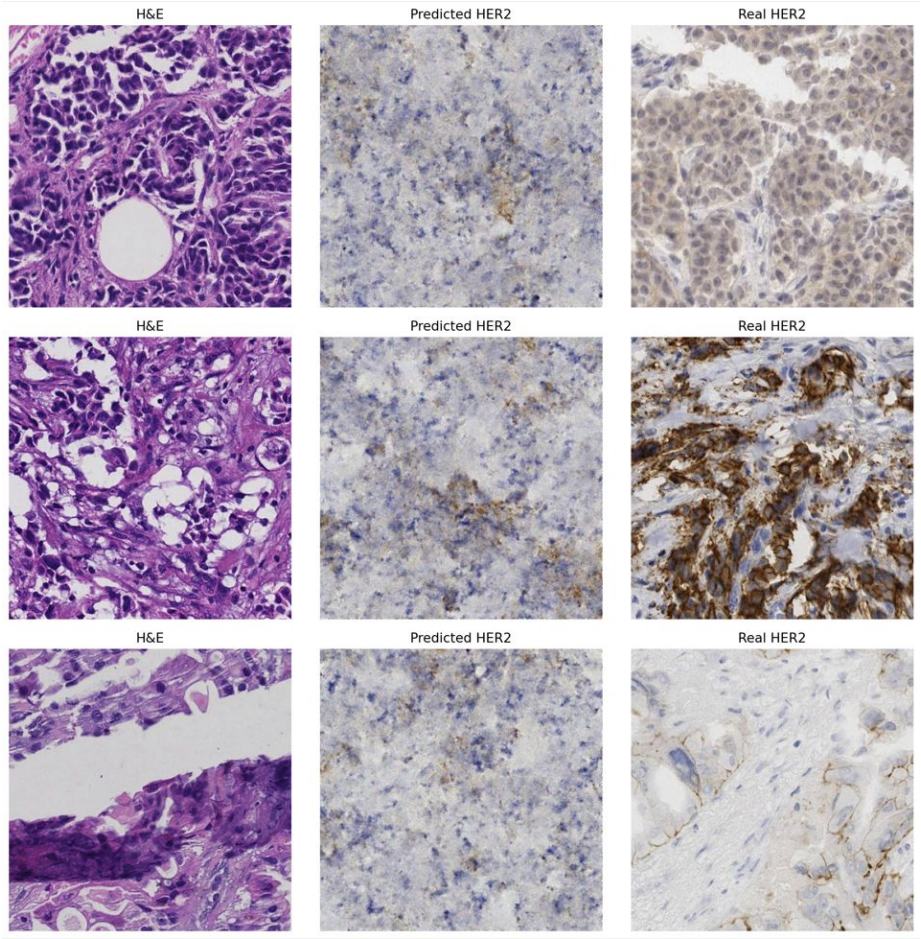


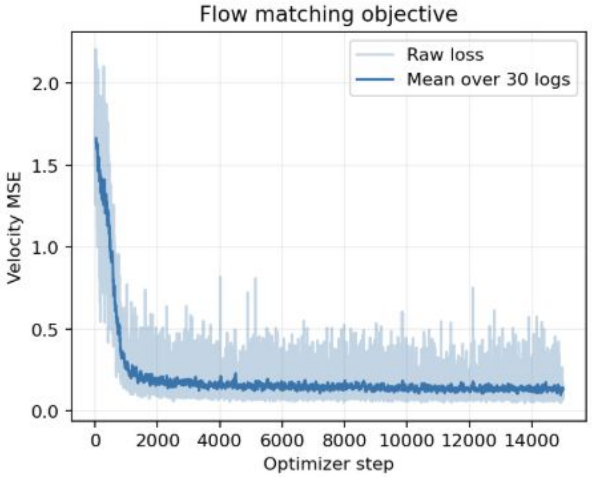


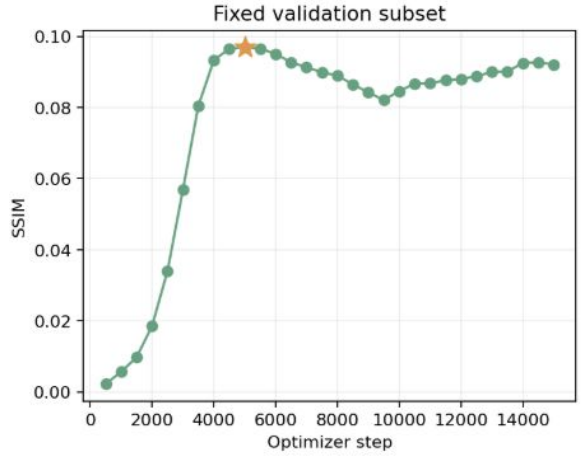


Conditioning diagnostic: Same initial noise with different H&E conditions produced similar HER2 predictions, indicating that baseline underuses the image condition.

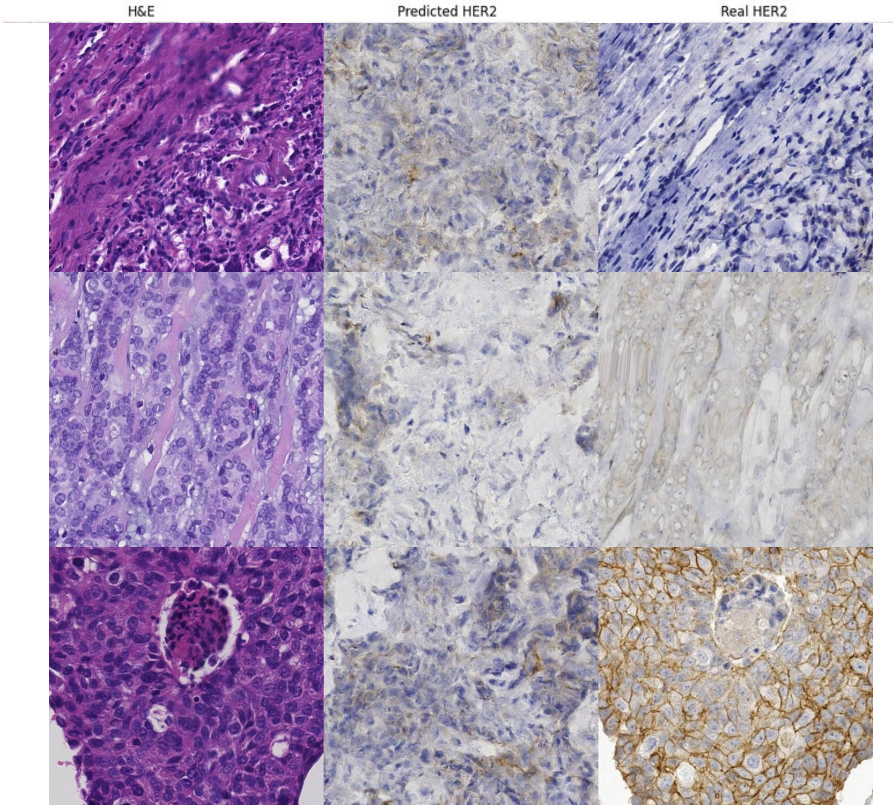
Variance-Preserving CFM MIST-HER2



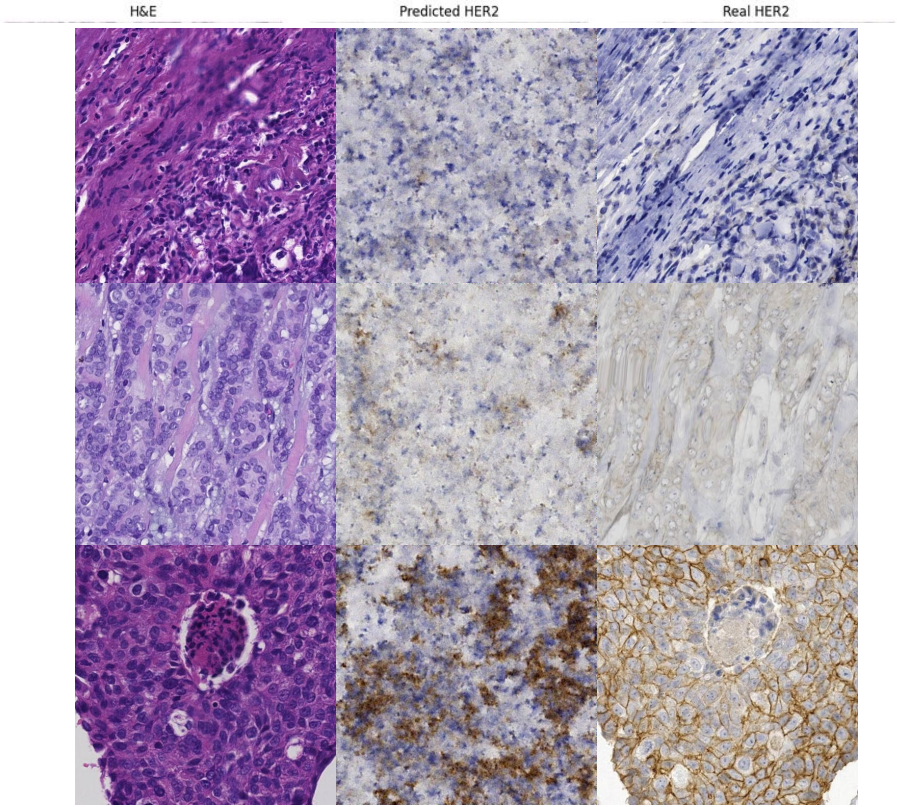




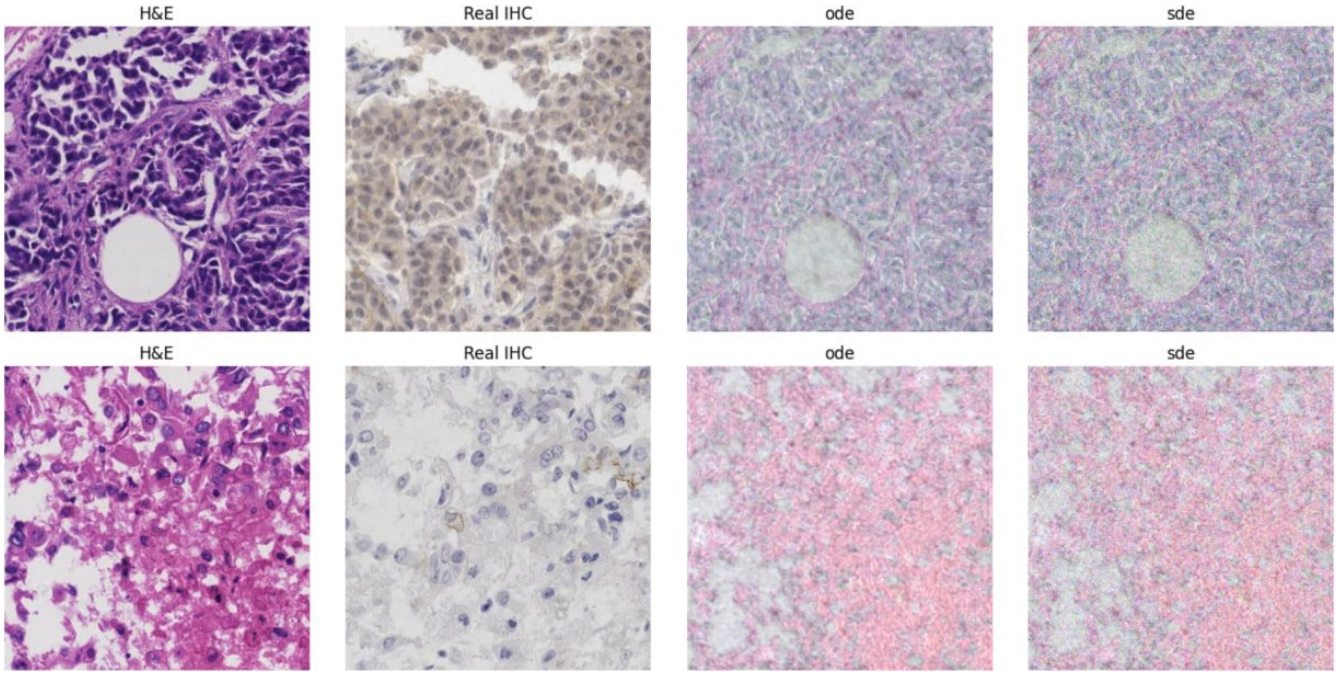
OT MIST-HER2



Blending with Phikon MIST-HER2

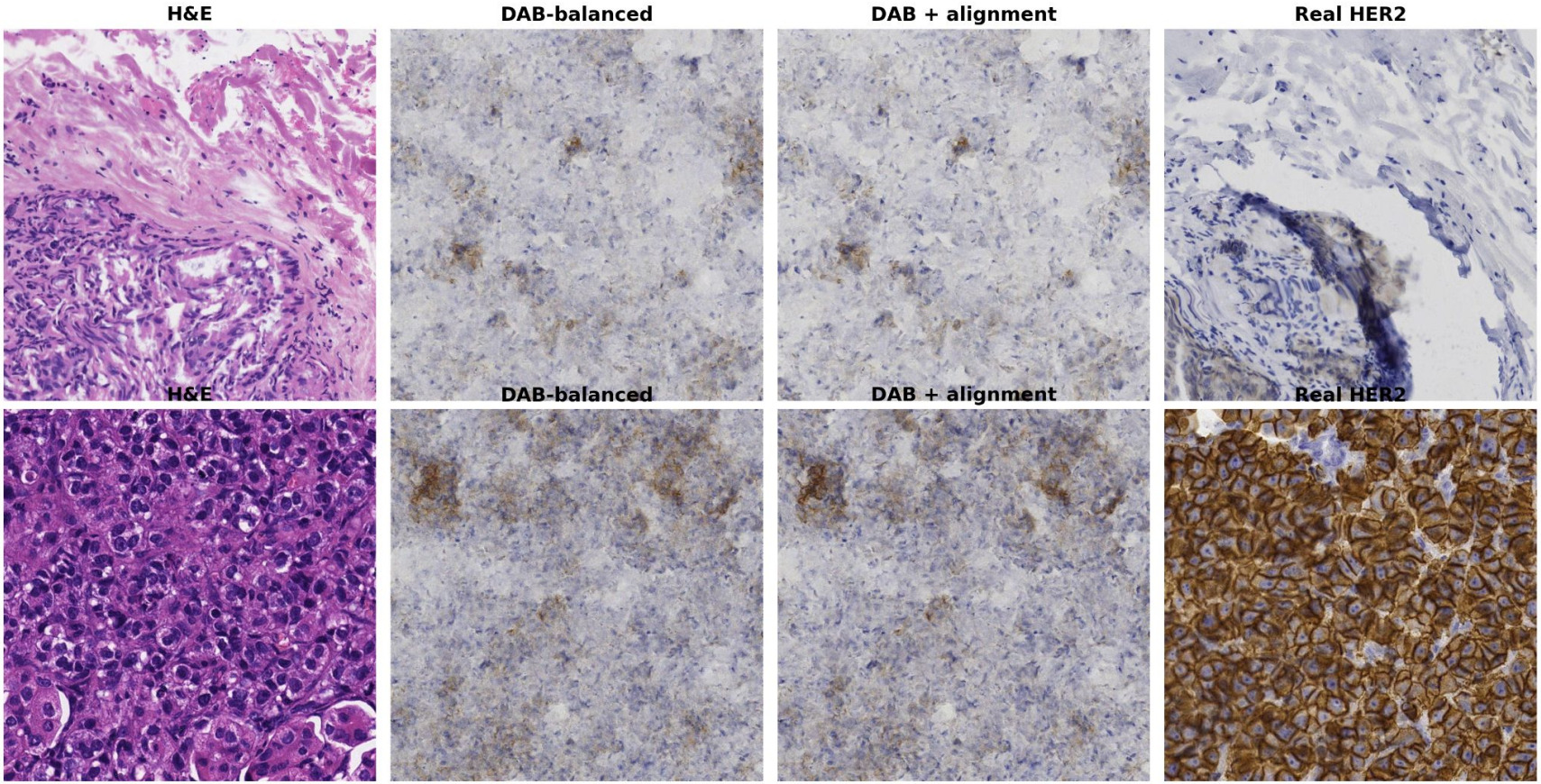


DirectFM baseline MIST-HER2



Diagnostic: Resulting we have something more like masks but not IHC

DAB-aware sampling: reasonable low-DAB reconstruction, but failure at high expression

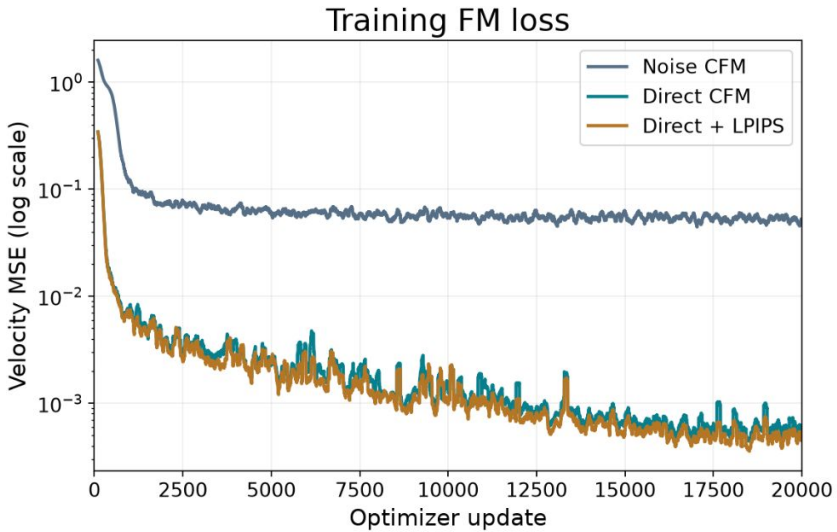


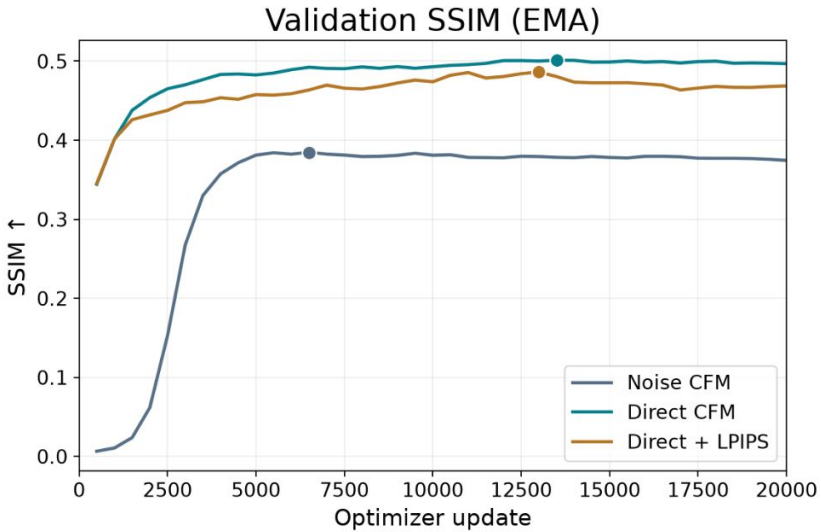
Preliminary Results MIST-HER2

Method	SSIM ↑	FID ↓	KID ×1000 ↓
CycleGAN [1]	0.1914	240.3	311.1
Pix2Pix [1]	0.1559	137.3	82.9
PyramidP2P [1]	0.2078	104.0	61.8
ASP [1]	0.2004	51.4	12.4
MDCL [2]	0.1810	44.4	7.5
HistDiST [2]	0.2059	36.9	3.8
UNIStainNet [3]	0.2290	34.5	2.2
CFM (Baseline)	0.1520	163	95
Variance-Preserving CFM	0.1447	367	469
CFM baseline (256×256 crops)	0.1700	222	255
CFM (HER2Match-train)	0.1438	217	211

HER2match

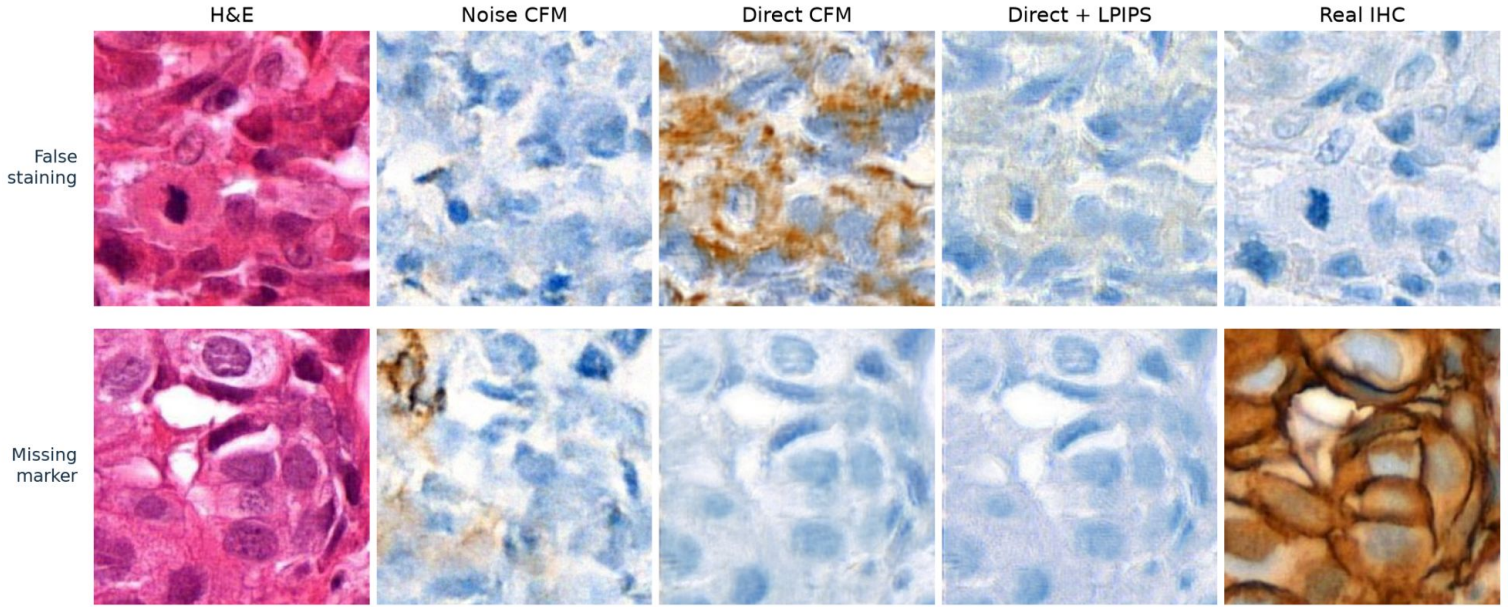
- Noise CFM: noise → IHC; Direct CFM: H&E → IHC; Direct + LPIPS adds a perceptual loss.
- Direct improves validation SSIM; next, we inspect whether the marker signal is also correct.





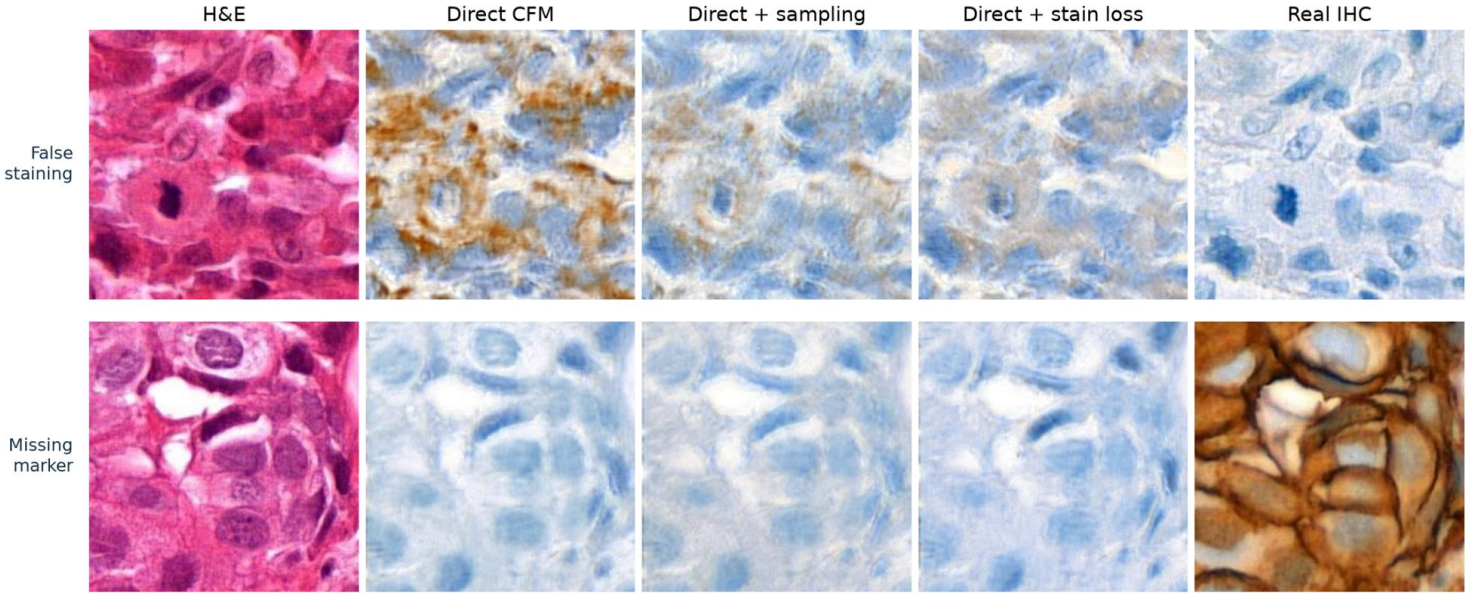
HER2match qualitative results

- Noise CFM → Direct CFM: SSIM 0.319 → 0.397; LPIPS 0.502 → 0.456.
- We tested Direct + LPIPS, then compared predictions with real IHC.
- **Problems:** false staining, missing marker signal, and loss of fine detail.



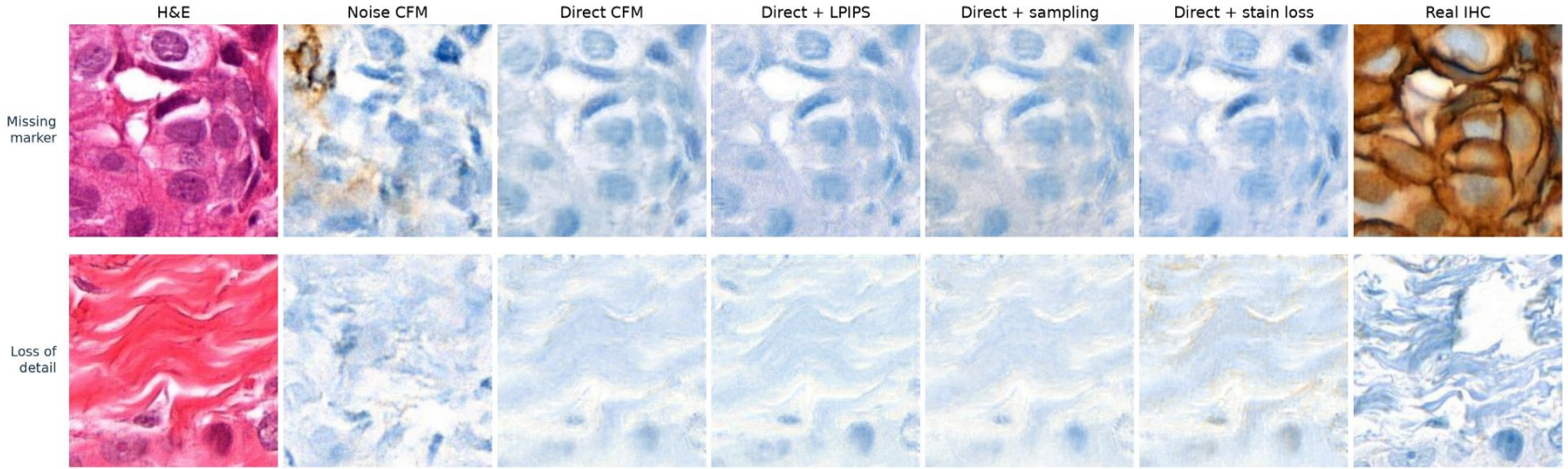
HER2match targeted fixes

- Sampling: expose the model to more moderately and strongly stained crops; retain negatives.
- Stain loss: penalize errors separately in stained and unstained tissue.
- **Results:** Both fixes reduced false staining, but did not eliminate it.



HER2match issues

- Strong-staining example: GT 73.35%; sampling 0.017%; stain loss 0.55%.
- Neither fix reduced the average missing-stain area.
- Fine structures remain smooth



HER2match final results

Method	SSIM ↑	FID ↓	PSNR↑
PyramidP2P [9]	**0.51**	67	20
ASP [9]	**0.51**	102	**21**
CSSP2P GAN [9]	0.43	**33**	18
CFM (Baseline)	0.32	221	17.7
DFM	0.40	151	18.9
DFM + lpips	0.39	131	18.9
DFM + sampling	0.41	158	19
DFM + stain loss	0.39	150	18.8

Key Findings and Insights

1. Pair alignment strongly affects performance

The baseline achieved higher SSIM on the well-aligned HER2match dataset than on MIST (0.33 vs. 0.15), indicating that imperfect H&E–IHC registration is a major challenge for pixel-level Flow Matching.

2. Direct transport is more effective on aligned pairs

On HER2match validation, direct H&E-to-HER2 CFM reached approximately 0.50 SSIM, compared with approximately 0.38 for noise-conditioned CFM. Starting directly from H&E may preserve tissue structure more effectively when image pairs are accurately aligned.

3. More complex variants did not automatically improve quality

Variance-Preserving CFM performed worse than the standard MIST baseline, especially in FID and KID.

4. Crop size creates a local–global quality trade-off

Smaller crops improved paired agreement but reduced target-distribution fidelity.

Limitations and next steps

What we achieved

Implemented conditional FM and variants.
Compared two datasets and diagnosed weak conditioning and staining errors.

What remains unresolved

Strong HER2 signal is still missed.
Some outputs contain extra brown staining.
Cross-dataset results do not isolate alignment.

Next steps

Isolate one change per experiment.
Use one validation and sampling protocol.
Evaluate positive/negative cases and DAB.

References

1. **Li F., Hu Z., Chen W., Kak A.** Adaptive Supervised PatchNCE Loss for Learning H&E-to-IHC Stain Translation with Inconsistent Groundtruth Image Pairs. MICCAI, 2023.
2. **Großkopf E. et al.** HistDiST: Histopathological Diffusion-based Stain Transfer. arXiv preprint, 2025.
3. **Saurav et al.** UNIStainNet: Foundation-Model-Guided Virtual Staining of H&E to IHC. arXiv preprint, 2026.
4. **Lipman Y., Chen R. T. Q., Ben-Hamu H., Nickel M., Le M.** Flow Matching for Generative Modeling. ICLR, 2023.
5. **Liu X., Gong C., Liu Q.** Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. ICLR, 2023.
6. **Tong A. et al.** Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport. Transactions on Machine Learning Research, 2024.
7. **Klöckner P., Teixeira J., Montezuma D., Cardoso J. S., Horlings H. M., Oliveira S. P.** GANs vs. Diffusion Models for Virtual Staining with the HER2match Dataset. arXiv preprint, 2025.
8. **HER2match dataset.** Paired H&E–HER2 tiles and whole-slide images. Zenodo, 2025.
9. **Teixeira J., Klöckner P., Montezuma D., Cesur M. E., Fraga J., Horlings H. M., Cardoso J. S., Oliveira S. P.**Leveraging Adversarial Learning for Pathological Fidelity in Virtual Staining. arXiv preprint, 2025.

Selected references and resources

Flow Matching

Lipman et al., Flow Matching for Generative Modeling, ICLR 2023

https://arxiv.org/abs/2210.02747

Rectified Flow

Liu et al., Flow Straight and Fast, 2023

https://arxiv.org/abs/2209.03003

OT-CFM

Tong et al., Minibatch Optimal Transport, TMLR 2024

https://arxiv.org/abs/2302.00482

MIST

Li et al., Adaptive Supervised PatchNCE, MICCAI 2023

https://arxiv.org/abs/2303.06193

HER2match

Klöckner et al., GANs vs. Diffusion Models, 2025

https://arxiv.org/abs/2506.18484

Dataset

HER2match tiles and whole-slide images

https://zenodo.org/records/15797050